

PATRICIA RIVERA

Ph.D

PATRICIA RIVERA PÉREZ

10 Cefiro,
México City, México 04700

(52) 5528017554
rivera.perezpatricia@gmail.com

Profile

I am a plant biologist with experience in the design and development of research projects focused on the area of botany and evolutionary biology. I have a background in molecular biology and anatomy, handling material and laboratory equipment as well as organizing, analyzing, and presenting the data obtained. At the same time, I have worked for several years as a statistics and plant biology teacher at Mexico's National University. I would like to combine my activities as a researcher with the formation of human resources.

Experience

Comparative Biology Department, Sciences Faculty, National University of México UNAM / Subject Teacher (Biostatistics/ Plant Biology)

AUGUST 2014 - PRESENT, MÉXICO CITY

I design and present theory and practice classes for undergraduate students on probability and general statistics as well as a course on angiosperm biology.

Tlalpan Departmental Unit for Youth and Gender Equality. Mexico City Government/ General Biology Teacher for the "Program of high-school student advisory"

DECEMBER 2011 - JULY 2011, MÉXICO CITY

I design weekly and monthly work plans for a general biology course at a high school level. I design didactic material, tests, and educational activities for adolescents.

National System of Researchers. National Council of Science and Technology (CONACyT) / Researcher's assistant

AUGUST 2010 - AUGUST 2011, MÉXICO CITY

I collect and process leaf and stem samples with conventional anatomical techniques: clearing, paraffin inclusion, staining, and mounting.

Publications

Rivera P., Villaseñor J.L., Terrazas T., Panero J.L. 2020. THE IMPORTANCE OF MEXICAN ASTERACEAE ON THE FAMILY PHYLOGENY. *Journal of Systematics and evolution*. Under revision

Rivera P., Terrazas T., Villaseñor J.L. Rojas-Leal A. 2019. LEAF ARCHITECTURE AND ANATOMY OF ASTERACEAE SPECIES IN A XEROPHYTIC SCRUB IN MEXICO CITY, MEXICO. *Acta Botánica Mexicana*. 126: e1515. DOI: 10.21829/abm126.2019.1515

Rivera P., Villaseñor J.L., Terrazas T. 2017. MESO- OR XEROMORPHIC? FOLIAR CHARACTERS OF ASTERACEAE IN A XERIC SCRUB OF MEXICO. *Botanical Studies*. DOI: 10.1186/s40529-017-0166-x

Rivera P., Villaseñor J.L., Terrazas T. 2013. EL APARATO ESTOMÁTICO DE ASTERACEAE Y SU RELACIÓN CON EL AUMENTO DE CO₂ ATMOSFÉRICO EN LA RESERVA ECOLÓGICA DEL PEDREGAL DE SAN ÁNGEL. *Revista Mexicana de Biodiversidad* 84: 499-508. DOI: 0.7550/rmb.30933

Courses Taught

Introduction to the use of tools for phylogenetic systematics ("Introducción al uso de herramientas para la sistemática filogenética"), Universidad Autónoma de Aguascalientes/ Instructor

OCTOBER 2019, AGUASCALIENTES, MÉXICO

Together with Dr. Carolina Granados Mendoza and Dr. Itzi Fragoso Martínez, design and present an introductory course on phylogenetic systematics for botanists with different backgrounds. This course was part of the activities of the XXI Mexican Congress of Botany

Education

Biology Institute, National Autonomous University of Mexico (UNAM) / Ph.D. in Biological Sciences

MARCH 2015 - SEPTEMBER 2019, MÉXICO CITY

Thesis: Polyploidy and its effect on foliar attributes on Asteraceae: a comparative biology approach ("La poliploidía y su efecto en los atributos foliares de Asteraceae: una aproximación de la biología comparada")

Supervisor: Dr. Teresa Terrazas

Biology Institute, National Autonomous University of Mexico (UNAM) / Master in Biological Sciences (Systematics)

AUGUST 2012 - FEBRUARY 2014, MÉXICO CITY

Thesis: Stomatal apparatus of Asteraceae in the Pedregal of San Angel Ecological Reserve, and its relation to cytology ("El aparato estomático de Asteraceae en la REPSA y su relación con citología")

Supervisor: Dr. Teresa Terrazas

Sciences Faculty, National Autonomous University of Mexico

(UNAM) / Bachelor in Biology

AUGUST 2005 - NOVEMBER 2010, MÉXICO CITY

Thesis: Stomatal apparatus of Asteraceae in the Pedregal of San Angel Ecological Reserve, Mexico ("El aparato estomático de Asteraceae en la Reserva Ecológica del Pedregal de San Angel, Mexico")

Supervisor: Dr. Teresa Terrazas

Update Courses

Biology Institute, UNAM / Selected topics in the study of morphologic adaptation in plants ("Temas en el estudio de la adaptación morfológica en plantas")

FEBRUARY 2017, MÉXICO CITY, MÉXICO

Instructors: Dr. Michael J. Donoghue, Yale University, USA. Dr. Erika Edwards, Brown University, USA. Dr. Mark E. Olson, UNAM, Mexico

9 hours

Centre de Restauració i Interpretació Paleontologic / The use of phylogenies in the study of macroevolution. Organized by Transmitting Science

SEPTEMBER 2015, ELS HOSTALET DE PIEROLA, BARCELONA, ESPAÑA

Instructor: Dr. Juan López Cantalapiedra, Museum für Naturkunde, Berlin, Germany

40 hours

Biology Institute, UNAM / Comparative biology, evo-devo, and paleobiology

SEPTEMBER 2013, MÉXICO CITY, MEXICO

Instructor: Michel Laurin, Centre de Recherches sur la Paléobiodiversité et les Paléoenvironnements, France.

30 hours

Biology Institute, UNAM / Linear and generalized linear models using R language ("Modelos lineales y lineales generalizados utilizando el lenguaje R")

NOVEMBER 2012, MÉXICO CITY, MEXICO

Instructor: Dr. Santiago Benitez-Vieyra, CONICET-Argentina.

40 hours

Participation in Conferences

Sociedad Botánica de México / XXI Congreso Mexicano de Botánica

OCTOBER 2019, AGUASCALIENTES, MEXICO

Moderator at the oral session: "OR26. Asteraceae"

Conference Speaker: Leaf variation in Asteraceae associated to polyploidy and growth form ("Variación foliar en Asteraceae asociada a la poliploidía y la forma de crecimiento"). Rivera P., Villaseñor J.L., Terrazas T.

Poster: Relationship between vegetation types and foliar morpho-anatomy of nine species of Asteraceae in México ("Relación entre tipos de vegetación y la morfo-anatomía foliar de nueve especies Asteraceae en México"). Rosas-Paz M., **Rivera P.**, Terrazas T. Villaseñor J.L.

Sociedad Latinoamericana de Botánica / XII Latin American Botany Congress

OCTOBER 2018, QUITO, ECUADOR

Organizer of the round table: "MR10. Avances en el estudio de las Asteráceas Latinoamericanas: una visión integradora"

Conference Speaker: Relation between leaf attributes and chromosome numbers in Asteraceae from the comparative biology perspective ("La relación entre atributos foliares y numeros cromosomicos en Asteraceae desde la perspectiva de la biología comparada"). **Rivera P.**, Villaseñor J.L., Terrazas T.

Sociedad Botánica de México / XX Congreso Mexicano de Botánica

SEPTEMBER 2016, MÉXICO CITY, MEXICO

Conference Speaker: Are the leaf characters of Asteraceae in the Mexico Basin xeromorphic? ("¿Son los caracteres foliares de Asteraceae de un matorral xerófilo en la Cuenca de México xeromórficos?"). **Rivera P.**, Terrazas T., Villaseñor J.L.

Botanical Society of America / Botany 2016

AUGUST 2016, SAVANNAH GEORGIA, USA

Poster: Comparative leaf anatomy of Asteraceae in a xerophytic scrub and its relation to cytogenetic characters. **Rivera P.**, Terrazas T., Villaseñor J.L.

Sociedad Botánica de México / XIX Congreso Mexicano de Botánica

OCTOBER 2013, TUXTLA GUTIERREZ, CHIAPAS, MEXICO

Poster: Anatomic characters of the leaves of some Asteraceae in a xeric environment ("Caracteres anatómico foliares de algunas Asteraceae en un ambiente xérico"). **Rivera P.**, Terrazas T.

Sociedad Botánica de México / XVIII Congreso Mexicano de Botánica

NOVEMBER 2010, GUADALAJARA, JALISCO, MEXICO

Poster: Modifications of the stomatic apparatus of Asteraceae of the Pedregal de San Angel Ecological Reserve due to atmospheric changes ("Modificaciones del aparato estomático de las Asteraceae de la Reserva Ecológica del Pedregal de San Ángel debido al cambio atmosférico"). **Rivera P.**, Terrazas T.

Awards

First place. Thesis contest on botanic topics celebrated in the frame of the XIX Mexican Conference of Botany.

Bachelor thesis, 2013.

Skills

I have used “R” for basic and multivariate statistics in my own research. As part of the Statistics course that I teach, I have taught several students how to implement different scripts to process their own data. I also have been learning Julia and Python on my own and have successfully modified some python pipelines to run my own analysis. I am independent, methodical, and disciplined and enjoy learning on my own.

I have experience using and installing open-source software like Ubuntu, Linux, and the office tools, graphical processors, and accessories required to work with these operative systems. I use Google Sheets and Google Drive to store, process, share, and update databases for Asteraceae leaf characters. Our database has more than 500 species and 87 characters and is in the process of being curated. I also have experience installing and running software for phylogenetic analysis and have experience using online platforms and remote servers.

As a senior student in the Structural Botany laboratory of UNAM's Biology Institute, I have trained seven undergraduate students in basic molecular biology and anatomical techniques.

I have successfully participated in the writing and submission of projects for funding within the University. As part of the following required to continue the funding, I have actively participated in the preparation of the project advancement reports.

Languages

Spanish (Native language)

English (Advanced)

Portuguese (Intermediate)